Auditing constrained geospatial allocation under noisy annual land-cover labels

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A Geospatial Kernel benchmark in Abu Dhabi

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This study is not an official Abu Dhabi planning forecast. Correspondence: Ning Zhou, Beijing Freedo Technology Co., Ltd.; zhouning@freedotech.com.

Abstract

Urban land-change models are often evaluated against annual labels whose reliability is not spatially uniform. We present an auditable protocol for testing constrained spatial allocation under that uncertainty, using Geospatial Kernel, the algorithmic core of a Geospatial World Model, as the focal execution layer. The Abu Dhabi public-data benchmark uses annual 100-m land-cover products from 2017–2024, three computational seeds, destination-correct multi-class change Figure of Merit (FoM), spatial-block bootstrap contrasts and explicit state–action–constraint traces. In a clean run of the fully resolved environment, strict FoM was 0.1961 for Geospatial Kernel, 0.1256 for the GeoSOS-derived FLUS-style ANN–CA console (author-modified build) and 0.1619 for GeoFM-LDN in 2023; after recursive two-step rollout in 2024 the values were 0.2529, 0.1782 and 0.2708, respectively. A dual-year confidence threshold retained only 25 of 4,500 observed changes in 2023 and 56 of 8,464 in 2024; all three FoM values were 0.000 in 2023 and the 2024 maximum was 0.0188. These filtered scores are label-quality and selection-effect diagnostics, not model-skill tests. The planning track is therefore reported as scenario stress testing, not official forecasting. The released public bundle, source, checkpoints, rasters, vector footprints and audit manifests support locked-environment reproduction; cross-stack variation is quantified rather than assumed absent. The protocol's principal contribution is an inspectable proposal–projection–writeback boundary for separating model behaviour from action and constraint semantics.

Highlights

- Geospatial Kernel separates transition proposals from constrained admission.

- Typed state–action–constraint traces make recursive updates auditable.
- Spatial-block uncertainty and paired model contrasts are reported explicitly.
- Confidence filters expose label selection effects; they are not validation tests.
- Planning outputs are scenario stress tests, not statutory Abu Dhabi forecasts.

Keywords

Geospatial World Model; land-cover change; constrained allocation; cellular automata; Abu Dhabi; scenario planning

Introduction

Land-cover change is a spatial process. Development demand is expressed as a quantity, but its consequences depend on where each new cell is placed relative to roads, existing built areas, ecological features and other land-cover classes. This coupling matters for urban planning because two maps with the same total built area can imply very different infrastructure requirements and environmental pressures. Cellular-automata and suitability-based models have long provided a way to distribute land demand across a map, including CLUE-S, SLEUTH, Dinamica EGO, FLUS, local-attribute urban CA models and more recent patch-generating or deep-learning CA models (Verburg et al., 2002; Silva and Clarke, 2002; Soares-Filho et al., 2002; Lau and Kam, 2005; Liu et al., 2017; Liang et al., 2021). Global Earth-observation products now make it possible to construct annual, spatially aligned inputs for cities that lack a harmonized local data archive (Brown et al., 2022; Gorelick et al., 2017).

Validation studies have also shown that agreement in unchanged cells can mask poor allocation of actual change. Quantity and allocation disagreement, destination-specific transition scores and explicit calibration/validation protocols are therefore preferable to a single overall-accuracy number (Pontius et al., 2008; Pontius and Millones, 2011; van Vliet et al., 2011, 2016). This distinction is central here because the benchmark uses noisy annual labels and tests both one-step allocation and recursive rollout.

The main technical difficulty is not producing another categorical raster. It is keeping the semantics of a planning action separate from the learned transition tendency. A model may predict that a cell is likely to become built, yet the resulting map can still violate a water exclusion, miss the requested class totals or become inconsistent when the next year is simulated from the model's own output. These failure modes are particularly important when the model is used to compare scenarios rather than to reproduce one observed transition. Latent-dynamics approaches such as GeoFM-LDN can represent continuous geospatial state and condition transitions on demand, whereas traditional ANN-plus-cellular-automata systems combine suitability with explicit neighbourhood evolution. Neither design, by itself, guarantees that action semantics, hard constraints, state persistence and evidence provenance are visible at the same execution boundary.

We address this systems problem by studying Geospatial Kernel, the core transition layer of a Geospatial World Model (GWM). The Kernel represents one simulation step as a typed state, a typed action, a transition proposal, a constraint projection and a state writeback. The learned proposal remains domain-specific; the runtime contract is domain-neutral. For Abu Dhabi land-cover planning, the proposal is a six-class probability cube, and the projection allocates only mutable cells until the action's feasible class totals are reached. This separation makes the planning operation inspectable: a reviewer can distinguish what the transition model preferred from what the constraints admitted.

We evaluate this design in a controlled public-data benchmark rather than treating the benchmark as an official data release. All candidates consume the same 100-m grid, common valid mask, annual class-count actions, hard exclusions and evaluator. The historical track uses observed target counts to isolate spatial allocation skill from demand error. The planning track starts from the observed 2024 state and applies three planner-supplied actions: moderate growth, green-priority growth and high outward growth. Abu Dhabi Plan 2030 provides an institutional planning context for discussing urban structure, but the numerical actions in this benchmark are not calibrated to that plan or to an official population forecast (Abu Dhabi Urban Planning Council, 2007). We ask four questions: whether the Kernel execution contract is auditable, whether its allocation skill is competitive, whether it produces feasible and spatially distinct allocations and which conclusions survive label and parameter sensitivity.

The paper makes three bounded contributions. First, it gives a concrete state–action–constraint formulation for using a geospatial kernel as an algorithmic core rather than as an opaque end-to-end predictor. Second, it defines an evaluation protocol that reports confidence-filter retention and selection effects rather than treating a high-confidence subset as a skill-validation set. Third, it turns future maps into explicit stress-test artefacts, including cumulative rasters and vectorized transition footprints, while preserving the distinction between public-data land cover and authoritative legal land use.

Methods

Portable audit protocol

The benchmark separates a reusable audit protocol from the Abu Dhabi adapter. For any city or geospatial domain, the protocol requires: (i) a declared class crosswalk and spatially stratified label-confidence tiers; (ii) persistence and minimum-change zero models; (iii) destination-correct multi-class change FoM with quantity and allocation diagnostics; (iv) spatial-block bootstrap intervals and paired model contrasts on a frozen evaluation mask; (v) an explicit check for structural-zero metrics before Pareto selection; (vi) a versioned objective table with direction, units and interpretation; and (vii) a state–action–constraint trace recording proposal, projection, hard-mask status and state writeback. The protocol also requires every model/year/seed record to report the number of predicted changed cells. A record with `predicted_change_pixels == 0` is marked as a degenerate output and reviewed alongside structural-zero metrics; it cannot by itself be interpreted as model superiority or real land-cover stability. The protocol reports domain validity separately from execution completeness and requires independent change validation before a categorical map is presented as an observed land-use product. A structural-zero metric is flagged operationally when its denominator is zero for every candidate under the declared action and constraint rules; for exact-count allocation, a conversion rate is also a structural zero when the protected/source rule makes the relevant source class monotonically non-decreasing, so no eligible source cells can contribute to that transition. Such a metric is reported descriptively and excluded from Pareto selection.

Abu Dhabi implementation: boundary, grid and temporal protocol

The study boundary is the polygon returned for OpenStreetMap relation R4479763, representing Abu Dhabi city. All rasters were aligned to a canonical 100-m grid in EPSG:32640 with 475 columns and 360 rows. A cell was included when its centre fell within the city polygon and when the common public-data mask indicated valid coverage. The benchmark profile records the source geometry, raster transform, dimensions and SHA-256 hashes. The effective evaluation mask contains 79,726 cells.

The observed sequence contains annual states for 2017–2024. Models were fit using the four transitions 2017→2018, 2018→2019, 2019→2020 and 2020→2021. A 2021→2022 transition was used for validation during model development. The historical test starts from 2022 and evaluates 2023 and 2024 in an open-loop rollout without observed-state writeback. The planning test starts from the observed 2024 state and recursively generates 2025–2031 under each scenario action.

Public data and class semantics

Dynamic World V1 provides annual scene-argmax labels and mean top-class probabilities (Brown et al., 2022). The nine Dynamic World labels were crosswalked to six canonical classes: water (source 0), woody vegetation (1), low vegetation (2, 4 and 5), wetland (3), built (6) and bare (7). Source class 8 was excluded. The resulting label is explicitly treated as land cover. No residential, commercial or industrial category is inferred from the raster labels.

Annual 64-dimensional AlphaEarth embeddings were spatially averaged to the canonical grid and locally L2-normalized. Monthly VIIRS night-time-light radiance was averaged by year. Copernicus DEM supplied elevation and a finite-difference slope derivative. OpenStreetMap road geometries were rasterized into distance-to-road and distance-to-major-road surfaces. ESA WorldCover 2021 supplied public water and wetland/mangrove constraint proxies (Zanaga et al., 2022). All dynamic and static layers were checked for exact CRS, transform, width and height agreement.

Geospatial Kernel state, action and transition proposal

Let $S_t(i)$ denote the six-class state at valid cell i, A_t the action for the interval $t \to t+1$, and X_t the aligned driver layers. The Kernel separates a proposal from a projection:

$$Q_t(i,c) = p_\theta\big(c \mid x_i(S_t, X_t)\big), \qquad S_{t+1} = \Pi_{\mathcal{C},A_t}(Q_t, S_t).$$

The feature vector x_i contains 25 values: six one-hot current-class indicators; twelve neighbourhood proportions, comprising 3×3 and 7×7 windows for each of the six classes; and seven continuous variables, namely normalized row and column coordinates, clipped elevation, clipped slope, log-transformed VIIRS radiance, log distance to roads and log distance to major roads. A histogram gradient-boosting classifier estimates the six-class probability vector. It uses learning rate 0.08, 120 iterations, at most 31 leaf nodes, minimum leaf size 40 and L2 regularization 1.0.

Training samples combine all changed valid cells and a 20% random sample of stable valid cells in each fit transition. The sample weight is the product of a change emphasis, $1 + 5\mathbf{1}(S_t \neq S_{t+1})$, and a confidence factor, $0.5 + \min(r_t, r_{t+1})$, where r is the Dynamic World mean top probability. The target years 2023 and 2024 are not used during fitting or model selection. Across the three seeds, each fitted model used 71,892–72,202 pixel rows and all six classes.

Constraint projection and state writeback

For a mutable source cell i of class s and target class c, the base allocation score is

$$q_{i,s\to c} = \log p_\theta(c \mid x_i) - \log p_\theta(s \mid x_i) + \lambda \rho^{(7)}_{i,c},$$

where $\rho^{(7)}_{i,c}$ is the 7×7 fraction of neighbouring cells currently in class c, and $\lambda = 0.35$. The algorithm computes class deficits and excesses from the action's feasible target counts, ranks all admissible source–target candidates by q, and changes cells in descending order until all deficits and excesses are zero. Permanent water and the protected wetland/ecological mask are held fixed. If the counts cannot be satisfied without changing a hard-exclusion cell, the step fails closed rather than silently violating the action.

The projected raster is written as the next `KernelState`. Every step records a state reference, action evidence, model version, parameter reference, projection status and diagnostic counts. The runtime checks domain identity, source-time consistency and action time advancement. This execution contract is the Geospatial Kernel's algorithmic contribution; the Abu Dhabi adapter supplies the land-cover-specific features, learner and constraint masks.

Baselines and common evaluator

The baseline was run through a GeoSOS-derived FLUS-style ANN–CA console (author-modified build; `paper-benchmark-flus-v1.1`) using seven continuous drivers: normalized row and column coordinates, elevation, slope, VIIRS radiance, distance to roads and distance to major roads. Its ANN suitability surface was passed to a cellular-automata allocation stage with the common class totals and public hard mask. The archived run used one 2021 training year, eight ANN hidden neurons, a 3 × 3 CA neighbourhood, neighbourhood strength 1.0, acceleration factor 0.1 and 1,000 iterations. Its frozen transition matrix protected water and wetland but did not prohibit built-to-non-built transitions. The algorithmic base is traceable to the GeoSOS team's public FLUS source release (Liu et al., 2017; geosimulation.cn); the accompanying source repository `FLUS_console_crossplatform` at commit `47e65b3` contains the author modifications used for this paper. Those modifications add `train/train-update` entry points and `FLUS_RANDOM_SEED` deterministic seeding for both ANN sampling and CA roulette draws, so the build is not the unmodified upstream executable. Version v1.1 changes only failure handling for invalid input reads; valid-input algorithmic paths and benchmark rasters are unchanged from v1. A 25-feature Kernel-matched input was then executed with absolute paths for seeds 31, 47 and 73 and is reported only as a platform-sensitive diagnostic. It shares the Kernel feature family, but not the learning target or projection semantics: the FLUS ANN estimates same-year label suitability $p(S_t \mid X_t)$, whereas the Kernel proposal is trained to estimate a next-state transition $p(S_{t+1} \mid S_t, X_t)$ before constrained projection. Matching features therefore does not match the training task. Two intermediate configurations were retained as mechanism diagnostics: 13 features (seven drivers plus current-class indicators) and 19 features (seven drivers plus neighbourhood fractions). The 25-feature run collapsed to zero change for macOS seeds 47 and 73 and for all three reviewer-provided Windows x86_64 seeds; the remaining macOS seed-31 run is not used as a valid estimator. The 19-feature mode produced changes for all three macOS seeds but underfilled the observed 4,500 and 8,464 changed cells in 2023 and 2024. Both modes are therefore diagnostic-only.

GeoFM-LDN (Geospatial Foundation-Model Latent Dynamics Network; the repository's former `paper58` path identifier) was implemented as a demand-conditioned residual latent-dynamics network on 64-channel AlphaEarth patches. Three 128-channel convolutions use dilation rates 1, 2 and 4, followed by a 1 × 1 projection to a 64-dimensional latent state. A 12-dimensional action vector (origin and target counts for six classes) is encoded by a two-layer MLP and broadcast across the patch before concatenation with the embedding. Group normalization and GELU activations are used in the residual blocks. A scikit-learn logistic-regression decoder maps the predicted latent embedding to six semantic classes. Training uses 64 × 64 patches, batch size 2, eight epochs, AdamW with learning rate 3×10^{-4} and weight decay 10^{-4}; the loss is the weighted sum of cosine embedding loss, 0.5 semantic cross-entropy, 0.1 demand-consistency loss and 0.2 hard-constraint consistency loss. The released checkpoints correspond to seeds 31, 47 and 73 and were trained in August 2026. The default reproduction loads these fixed checkpoints rather than retraining; the runner exposes retraining separately. Fixed checkpoints do not remove the decoding runtime boundary: categorical output still depends on the locked Torch and scikit-learn stack. Same-stack reproduction is the declared reference; no independent cross-stack GeoFM-LDN raster comparison is claimed, and changes between pre-lock and locked-environment outputs are not attributed to CPU architecture or to one library. During allocation, GeoFM-LDN directly calls the Geospatial Kernel `allocate_action` routine, so both methods share the same constrained projection and differ primarily in their proposal model. The comparison is therefore a proposal/pipeline comparison, not a comparison of two independent projection architectures. The name describes this benchmark implementation and does not attribute an external publication or performance claim.

The common evaluator computes actual class counts, normalized demand total variation, hard-mask violations, strict multi-class change FoM, binary change FoM, change F1, overall accuracy and macro-F1 for historical runs. The strict FoM counts a hit only when the predicted destination class equals the observed destination class; wrong destination changes are retained in the denominator. It also generates persistence and minimum-change zero controls, spatial-block bootstrap intervals and paired model-difference intervals for strict FoM, overall accuracy and macro-F1. For planning runs it additionally computes ecological conversion, new-built neighbourhood fraction, new-built component counts, the union of 2024 built and newly built components, all-final-built component counts, a 500-m leapfrog rate, distances to roads and prior built cells, infrastructure proxy distance and built retirement. Pareto membership is calculated within each scenario over three release objectives: major-road distance, prior-built distance and union-built component density. Ecological conversion is retained as a descriptive public-data pressure proxy because it is structurally zero for the two exact-count allocators and therefore has no discriminating power in their comparison. Vegetation gain is a scenario input and remains descriptive, not an objective.

Scenario actions and uncertainty

The moderate-growth, green-priority-growth and high-outward-growth actions are stored in the tracked `planning_scenarios_public_2025_2031.json` manifest; the legacy `compact` path identifier is retained only for artifact compatibility. Each action supplies feasible class totals for 2025–2031. The 2031 totals are an explicit extension of the previous annual difference and are not derived from an official population, housing or infrastructure forecast. Future exogenous rasters are held at their 2024 values. The green-priority action is not a water-budget model. Each candidate is run at seeds 31, 47 and 73, and planning tables report arithmetic means and population standard deviations. No p-values are reported because the three seeds describe computational variability, not independent field samples. The primary Pareto comparison is within each scenario, contrasting the three models under the same action; a global nine-candidate frontier is retained only as lineage. Sensitivity variants remove or add objective dimensions on the same released rasters. The objective set evolved through four review-driven revisions and was not preregistered: v1 mixed feasibility, action and outcome terms; v2 was not released; v3 used structurally problematic outcomes; and v4 still included scenario-supplied vegetation gain and all-built fragmentation. The current release is v6: road access, prior-built distance and union-built component density. Ecological conversion is reported as a diagnostic rather than a Pareto objective because it is structurally zero for the exact-count Kernel and GeoFM-LDN allocations. We treat v6 as a release objective set rather than a prespecified planning preference, and treat its sensitivity results as robustness diagnostics.

Change polygons and reproducibility

For each model–scenario pair, annual and cumulative differences from the 2024 raster are extracted as changed 100-m cells and dissolved into GeoPackage layers. The public delivery manifest covers 45 ensemble rasters for 2027–2031 and nine vector packages. The audit script reports `INCOMPLETE_INPUTS` or `FAIL` when source rasters, reports or vectors are absent; with the released public bundle, the input audit passed all alignment gates (with a Dynamic World label-noise warning) and the output audit passed 276 historical and planning predictions with zero failures. The vector layers preserve raster-cell geometry and should not be interpreted as cadastral parcels.

Results

A common benchmark separates allocation skill from demand assumptions

The benchmark covers the Abu Dhabi city polygon represented by OpenStreetMap relation R4479763. The canonical grid is a 475 × 360 lattice in EPSG:32640 at 100-m resolution. After the common coverage and alignment checks, 79,726 cells were used for scoring. Each cell represents 1 ha. Annual Dynamic World observations from 2017 to 2024 were harmonized to six land-cover classes: water, woody vegetation, low vegetation, wetland, built and bare. AlphaEarth annual embeddings, VIIRS night-time lights, Copernicus DEM derivatives and OpenStreetMap road distances supplied transition drivers. ESA WorldCover and public OpenStreetMap geometries supplied water, wetland and infrastructure exclusion proxies.

The comparison shared the canonical grid, origin states, target actions, hard masks and evaluator, but the public-data run was not information-balanced. The original FLUS-style control used seven continuous drivers in its external ANN suitability model, whereas Geospatial Kernel used 25 features including current-class indicators and neighbourhood proportions; GeoFM-LDN used AlphaEarth latent embeddings and the Kernel allocator. We therefore report the original result as an unmatched-input pipeline comparison, not evidence that one learner is intrinsically superior. The additional 25-feature FLUS run is a feature-space diagnostic only, because it still learns same-year suitability rather than next-state transitions and does not use the Kernel projection implementation. Across six platform–seed runs, five produced zero-change outputs through current-class identity leakage. The archived FLUS console is a Mach-O arm64 executable rebuilt from the public GeoSOS source base with the author-modified `train`/`train-update` entry points and `FLUS_RANDOM_SEED` seeding patch; its exact source is archived at `FLUS_console_crossplatform` commit `deb0a54`. It is not asserted to be bitwise equivalent to an unmodified upstream build. All headline and diagnostic runs used seeds 31, 47 and 73.

The execution trace makes the distinction between proposal and admission explicit. A Kernel step records the source state, action, probability-cube proposal, projected state, model identity, input evidence and next-state reference. The runtime checks that the action advances the source time and that the projected raster becomes the next state. This is the operational meaning of the Kernel in this study: not a claim that all GWM domains share the same learner, but a claim that they can share an auditable execution contract. The shared inputs and execution boundary are summarized in Fig. 1.

Historical and planning results

The historical track uses observed target class totals to isolate spatial allocation from demand estimation. The primary metric is strict multi-class change FoM: a hit requires the predicted destination class to equal the observed destination class, while wrong destination changes remain in the denominator (Pontius et al., 2008). Overall accuracy, macro-F1, change F1 and binary FoM are secondary diagnostics. Uncertainty is estimated with an 8 × 8-pixel spatial-block bootstrap, and model contrasts are computed from the same resampled blocks for each frozen seed. Persistence and random minimum-change allocation are explicit zero-model controls.

The planning track starts from the observed 2024 state and applies three synthetic class-count actions through 2031. The release objective set is version 6. It uses mean distance of new built cells to major roads, mean distance to pre-existing built cells and connected components in the union of 2024 built and newly built cells per 1,000 valid cells. These are public-data proxies. Ecological conversion is retained as a descriptive diagnostic, not an objective, because it is structurally zero for the two exact-count allocators. Vegetation gain is excluded because it is supplied directly by the scenario action, and all-final- built component density is reported only as a diagnostic because it can be changed by built-cell retirement. No weighted composite score is used. Because the objective set changed across the review rounds, we disclose the lineage: v1 included seven access, ecological, neighbourhood and gain terms; v2 proposed roads, prior-built distance, fragmentation and leapfrog rate but was not run; v3 used ecological conversion, roads, leapfrog rate and built retirement; and v4 combined roads, prior-built distance, all-built fragmentation and vegetation gain. The current v6 release replaces v5 after the review identified the structural-zero ecological objective and the semantic inversion of new-only fragmentation. Sensitivity sets are recomputed on the same rasters and reported as robustness analyses, not as additional preregistered claims. These are measurable public-data proxies rather than claims about welfare, equity or statutory zoning.

The released public-data bundle supports a complete re-scoring of the current historical rasters and a re-compilation of the 2025–2031 planning rasters. The strict multi-class transition FoM (mean across three seeds) was 0.1256, 0.1961 and 0.1619 for the FLUS-style console, Geospatial Kernel and GeoFM-LDN in 2023, and 0.1782, 0.2529 and 0.2708 in 2024, respectively. Under this unmatched public-data pipeline, Geospatial Kernel had the largest 2023 value, whereas GeoFM-LDN had the largest 2024 value after recursive two-step rollout; no single model dominated both horizons. These rankings are conditional on the different inputs, learners and shared evaluation projection.

Table 1 | Historical allocation scores. Values are means across the three frozen computational seeds; the strict multi-class change FoM is the primary metric and the remaining columns are diagnostics. The 25-feature FLUS run is excluded from this table because it is not a comparable estimator. Across six platform–seed runs, five collapsed to zero-change outputs through current-class identity leakage, so the non-collapsed macOS seed-31 result is retained only in the diagnostic archive. Persistence and random minimum-change are explicit zero-model controls rather than learned models.

Target year	FLUS-style (7)	Kernel	GeoFM-LDN	Persistence	Random minimum-change
2023 (1-step)	0.1256	**0.1961**	0.1619	0.0000	0.0253
2024 (2-step open-loop)	0.1782	0.2529	**0.2708**	0.0000	0.0605

Spatial-block bootstrap model contrasts support these differences. For 2023, Kernel minus the FLUS-style control was 0.0703 [0.0558, 0.0866] and GeoFM-LDN minus Kernel was -0.0341 [-0.0479, -0.0209]. For 2024, GeoFM-LDN minus Kernel was 0.0177 [0.0033, 0.0314]. These are conditional pipeline contrasts on public labels, not independent-sample significance tests. The persistence control had higher overall accuracy than every learned candidate (0.9436 in 2023 and 0.8938 in 2024), illustrating why change-specific metrics are necessary. Fig. 2 shows the four historical metrics alongside the persistence and random minimum-change controls; main-model bars use three-seed population standard deviations, whereas the paired spatial-block intervals above are the uncertainty summaries used for model contrasts. The 25- and 19-feature FLUS diagnostics are reported in the machine-readable archive and Supplementary Table S3.

The reviewer-requested FLUS input-dimension analysis was also completed. The 25-feature run is not a valid point estimator: macOS seeds 47 and 73 and all three reviewer-provided Windows x86_64 seeds produced zero-change outputs, whereas macOS seed 31 was the sole non-collapsed run. The six-run pattern is therefore platform-sensitive and is retained as an identity-leakage diagnostic, not as a Kernel contrast. The 19-feature diagnostic was run for all three macOS seeds. Its strict FoM ranged from 0.1071 to 0.1617 in 2023 and from 0.0872 to 0.1386 in 2024; predicted changes ranged from 1,702 to 2,417 and from 1,805 to 2,499 cells, respectively, compared with 4,500 and 8,464 observed changes. Demand total variation ranged from 0.0153 to 0.0305 in 2023 and from 0.0615 to 0.0767 in 2024. These results show why matching feature dimensions cannot make the tasks equivalent: FLUS learns static suitability $p(S_t \mid X_t)$, while the Kernel learns next-state transitions and then projects them against action deficits and hard constraints.

The confidence sensitivity is a label-quality and selection-effect diagnostic, not an independent validation set or a model-skill test. Requiring both the origin and target Dynamic World mean top probability to exceed 0.5 retains only 25 of 4,500 observed changes (0.56%) in 2023 and 56 of 8,464 (0.66%) in 2024. Strict FoM is 0.000 for all three models in 2023; in 2024 the means are 0.0040 for the FLUS-style console, 0.0055 for Geospatial Kernel and 0.0188 for GeoFM-LDN. Screening only the confidence of the year immediately preceding the target avoids filtering on target-year confidence, but still retains only 129 (2.87%) and 93 (1.10%) observed changes in 2023 and 2024, respectively. Its 2023 FoM values remain 0.000; its 2024 means are 0.0040, 0.0041 and 0.0120 for the FLUS-style console, Kernel and GeoFM-LDN. The two filters therefore describe label selection rather than establishing or rescuing model skill. Dynamic World also reports built pixels increasing from 9,359 in 2022 to 15,598 in 2024 (67%), while the median one-year reversion fraction is 0.3611 and the mean fraction of cells below confidence 0.5 is 0.5497. Test-year change counts are approximately twice the training-transition counts. Because the same grid cells recur across years and the Kernel includes coordinate features, a training–test memory effect cannot be excluded; the reported scores should therefore be interpreted as conditional agreement with the public product.

The reference reproduction environment is macOS arm64 with Python 3.11 and scikit-learn 1.9.0. The archived Linux arm64 rerun uses Python 3.12 and scikit-learn 1.8.0, so it measures a cross-stack boundary rather than an architecture-only effect. Across six rasters (2023 and 2024 for each seed), it differs from the current reference by 498–1,147 cells (0.62–1.44% of valid cells), with strict-FoM deltas from -0.0023 to 0.0010. The reviewer-provided Windows FLUS rerun additionally showed 2–4% differing cells from the macOS seven-driver outputs, consistent with the platform-dependent C `rand()/srand()` stream. The repository does not contain the reviewer's Windows raster files, so those values are explicitly external verification rather than a local replay. The archived comparison report records the exact paths and environment fields for the arm64 pair. This documents measured cross-stack sensitivity, not universal bitwise identity.

The planning compiler compares the three models within each of the three scenarios. At 2031, the FLUS-style control has union-built component densities of 2.48–3.27 per 1,000 valid cells, the Kernel 2.92–4.28 and GeoFM-LDN 3.21–4.35. The Kernel is closer to major roads and prior built cells, whereas the union metric shows that this edge-filling pattern does not automatically imply lower overall fragmentation than the FLUS-style allocations. Under the release objective set, the within-scenario frontiers contain FLUS-style and Kernel candidates in all three scenarios; GeoFM-LDN is dominated in each scenario. In the preceding v5 formulation, GeoFM-LDN could appear non-dominated only because its ecological-conversion value was the same structural zero as the exact-count Kernel; that zero is not a model advantage. Ecological conversion is therefore retained only as a diagnostic in v6. All sensitivity variants retain the same FLUS-style/Kernel two-model frontier. These statements are conditional on public proxy layers and synthetic actions, not model superiority claims. The corresponding objective profiles and diagnostic trade-offs are shown in Fig. 3. The 2031 majority-vote ensemble L1 demand errors were 8, 44 and 42 pixels for the Kernel's moderate, green-priority and high-outward scenarios, respectively; the corresponding GeoFM-LDN errors were 1,580, 1,524 and 1,148 pixels and the FLUS-style errors were 2,504, 2,888 and 2,568 pixels. Seed-level projections meet their feasible class totals, but majority voting can produce a different raster and therefore a non-zero ensemble error. All 276 historical and planning prediction records passed the raster audit. The resulting 2031 ensemble rasters and their dissolved cell-change footprints are shown in Fig. 5; they are scenario-conditioned land-cover maps, not parcel boundaries or statutory zoning maps.

Table 2 | 2031 within-scenario planning objectives. Values are means across three seeds. Distances are metres, union-built component density is the number of connected components in the combined 2024-built and newly built footprint per 1,000 valid cells, and ecological conversion is a descriptive fraction of new-built cells whose origin class was woody, low vegetation or wetland. A star denotes membership in the primary within-scenario Pareto frontier.

Model	Scenario	Major-road distance	Prior-built distance	Union-built components/1,000	Ecological conversion (diagnostic)	Frontier
FLUS-style ANN–CA	Moderate	446.1	231.6	3.035	0.0168	*
Geospatial Kernel	Moderate	334.7	116.7	3.943	0.0000	*
GeoFM-LDN	Moderate	475.9	239.7	4.056	0.0000	
FLUS-style ANN–CA	Green-priority	443.7	212.7	3.265	0.0177	*
Geospatial Kernel	Green-priority	323.1	111.0	4.277	0.0000	*
GeoFM-LDN	Green-priority	452.1	222.3	4.348	0.0000	
FLUS-style ANN–CA	High outward	443.5	255.3	2.484	0.0131	*
Geospatial Kernel	High outward	365.5	139.8	2.923	0.0000	*
GeoFM-LDN	High outward	507.5	263.0	3.207	0.0000	

Sensitivity shows a stable but not fully explained neighbourhood contribution

The base Kernel allocation score adds a 7×7 target-class neighbourhood fraction with weight 0.35. The existing sensitivity and mechanism artefacts are retained as diagnostic controls; they do not identify a causal effect because the proposal features and allocator both contain spatial context. The neighbourhood term is therefore interpreted as evidence about the execution mechanism, not as an independently estimated planning preference. Figure 4A shows the proposal controls, while Fig. 4B–D reports runtime and planning controls.

In the post-hoc neighbourhood-weight sensitivity, mean strict FoM ranged from 0.19570 to 0.19609 in 2023 and from 0.25273 to 0.25304 in 2024; the complete strict-evaluator scan is provided in Supplementary Table S2. These changes are small relative to the model contrasts and are not interpreted causally. Objective-set sensitivity on the planning rasters retained the FLUS-style and Kernel candidates in all scenarios. GeoFM-LDN was dominated in all three scenarios after the structurally zero ecological-conversion term was removed from the release objectives.

Mechanism controls separate learned proposal from runtime projection

Mechanism controls separate the learned proposal from runtime projection, hard constraints and recursive writeback. They are not causal policy experiments. Deleting the action reduces mean strict FoM by 0.1124 in 2023 and 0.1699 in 2024, whereas deleting the hard constraint increases it by 0.014 and 0.011, respectively. The former is expected because the historical task supplies oracle target counts; it shows that much of the measured skill belongs to the action-conditioned allocator rather than to a demand forecast. The latter is an important warning: allowing changes inside the exclusion mask can improve agreement with noisy labels while violating the planning contract. The controls therefore support an execution-level interpretation but do not prove that either component causes an urban outcome.

Discussion

This study positions Geospatial Kernel as a candidate execution layer for constrained spatial reasoning and, more broadly, proposes an audit protocol for noisy annual labels. The defensible contribution is that a transition proposal and a planning admission rule can be represented and inspected separately. Under the unmatched public-data pipeline, Geospatial Kernel has the largest strict transition FoM at the one-step horizon, while GeoFM-LDN has the largest value after recursive two-step rollout. The dual-year confidence filter retains only 0.56% and 0.66% of observed changes in 2023 and 2024, respectively; the preceding-year-only variant retains 2.87% and 1.10%. These subsets are not skill tests, but their event-dependent selection shows why full-grid rankings cannot be interpreted as validated urban-change skill. We therefore treat the planning frontier as conditional on public proxy objectives and synthetic actions.

The evidence remains bounded. The six classes are remote-sensing land cover, not statutory residential, commercial or industrial land use. Dynamic World labels show substantial annual uncertainty and apparent turnover; no independent 2023–2024 change product or observed 2025–2031 labels was available. Public roads, wetland and protected-area layers are proxies, future drivers are held at 2024, and the green-priority action has no water-budget constraint. The FLUS-style control is an Apple-Silicon binary rebuilt from the public GeoSOS source base with the author's `train/train-update` and deterministic-seeding patches; its headline comparison uses seven drivers. The 25-feature run is a platform-sensitive identity-leakage diagnostic: five of six platform–seed runs collapsed to zero change, so the non-collapsed macOS seed-31 result is not used as a comparative estimate. These public-data stress tests therefore do not establish causal planning effects or a validated Abu Dhabi forecast. An authoritative deployment would replace public labels and masks with locally approved land-use, infrastructure, ecological and development-demand records while retaining the same execution contract.

Data availability

The public-data benchmark manifests, protocols, reports and generated delivery artefacts are organized under `benchmarks/abu_dhabi_land_use_v1/` in the accompanying repository. The main evidence files are:

- `comparison_report_current.json` and `comparison_report_current.md` for historical metrics when a data-complete rerun is available;
- `planning_comparison_report_public_2025_2031_current.json` and its Markdown rendering for 2031 planning objectives;
- `planning_scenario_report_public_2025_2031_current.json` for per-seed rollout traces;
- `planning_public_2025_2031_delivery_manifest_current.json` for raster and vector delivery;
- `output_audit.json` for the current raster audit; `output_audit_reproducible.json` is a compatibility copy produced by the same run;
- `data_audit.json`, `protocol.json`, `gee_input_manifest.json` and `osm_input_manifest.json` for data provenance;
- The released bundle includes the aligned 2017–2024 public rasters, historical seed and ensemble prediction rasters, 2025–2031 planning seed and ensemble rasters, nine GeoPackages of dissolved change footprints, the mechanism-ablation report, model checkpoints, and the SHA-256 manifest. Text hashes use canonical LF line endings, and the byte-length check applies the same normalization on text records, so a Windows checkout with `core.autocrlf=true` can pass the manifest gate. The locked macOS arm64 execution environment remains the reference for model outputs. An archived arm64 Linux rerun of the six historical Kernel rasters differed from the macOS reference by 498–1,147 valid cells (0.62%–1.44%), with strict-FoM deltas from -0.0023 to +0.0010; these cross-stack differences are quantified rather than treated as categorical identity. Reviewer-provided Windows x86_64 values are separately labelled as external verification because the corresponding rasters are not archived locally. No private database address, credential or client-only service

is included. Authoritative Abu Dhabi land-use and independent change-validation data remain outside this study.

Code availability

The analysis scripts, a vendored Geospatial Kernel runtime snapshot, benchmark protocol, model runners and figure-generation code are maintained in the accompanying repository. Principal entry points are listed separately to keep the manuscript readable:

- `run_geospatial_kernel.py` for the Kernel historical run;
- `data_agent/uwm/geospatial_kernel/runtime.py` for the vendored runtime snapshot;
- `render_nature_figures.py` for publication artwork.
- `reproducibility/reproduce.py` for the end-to-end rerun;
- `reproducibility_check.py` and `audit_outputs.py` for fail-closed integrity checks.

GeoFM-LDN source and the three fixed checkpoints are included under `benchmarks/abu_dhabi_land_use_v1/external/` and `artifacts/predictions/`. Earlier internal reference outputs, produced before the dependency lock was completed, could not be regenerated under the resolved environment and were superseded by the locked-environment rerun reported here. Historical strict-FoM differences were at approximately the 0.001 level; discrete multi-year planning ensemble demand errors changed more substantially and are reported here only from the locked-environment rerun. The prior outputs remain Git-history lineage, not current evidence. The vendored FLUS executable is a macOS arm64 binary rebuilt from the GeoSOS public FLUS source base. The author-modified source, including `train-update` and `FLUS_RANDOM_SEED`, is archived at `https://github.com/zhouning/FLUS_console_crossplatform`, tag `paper-benchmark-flus-v1.1` at commit `47e65b3` (the GeoSOS source release and Liu et al. (2017) are cited above). Rebuilding the preceding v1 commit (`deb0a54`) on macOS arm64 produced SHA-256 `9839ce50950442d4ef49e4d2129a1ba27735e1955c2d4975ae51067c1c3c9964`, identical to the vendored executable. Version v1.1 changes only invalid-input failure handling and leaves valid-input algorithmic paths unchanged; the normal-output equivalence check is archived with the source release. The tagged author-modified source can also be built on Linux and Windows using the repository instructions. `FLUS_RANDOM_SEED` guarantees determinism only within a platform and build environment because ANN sampling and CA roulette draws use platform-defined C `rand()/srand()` streams. The reviewer-provided Windows rerun differed from macOS by approximately 2–4% of valid categorical cells for the seven-driver control, and all three Windows 25-feature runs collapsed to zero change. Windows execution requires the benchmark and working directories to use pure ASCII paths; saving configuration files as UTF-8 alone does not repair GDAL path decoding. A compatible local build is required to execute the FLUS track outside macOS arm64 and will produce a different random stream. The macOS binary SHA statement is an author self-check. As an upstream-only fallback, the public FLUS CA source can be paired with an open Python ANN implementation, but the unmodified upstream CA uses time seeding rather than `FLUS_RANDOM_SEED`. The corresponding-author e-mail is `zhouning@freedotech.com`; an archival DOI should be added after a persistent public release is created.

Declarations

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Competing interests. Ning Zhou is employed by Beijing Freedo Technology Co., Ltd., which develops geospatial-world-model software. The author declares no other competing financial or personal interests.

CRediT author statement. Ning Zhou: Conceptualization, methodology, software, data curation, formal analysis, visualization, writing—original draft, writing—review and editing.

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Ethics statement. Not applicable; the study used geospatial raster, vector and derived public-data products and did not involve human or animal participants.

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Figure legends

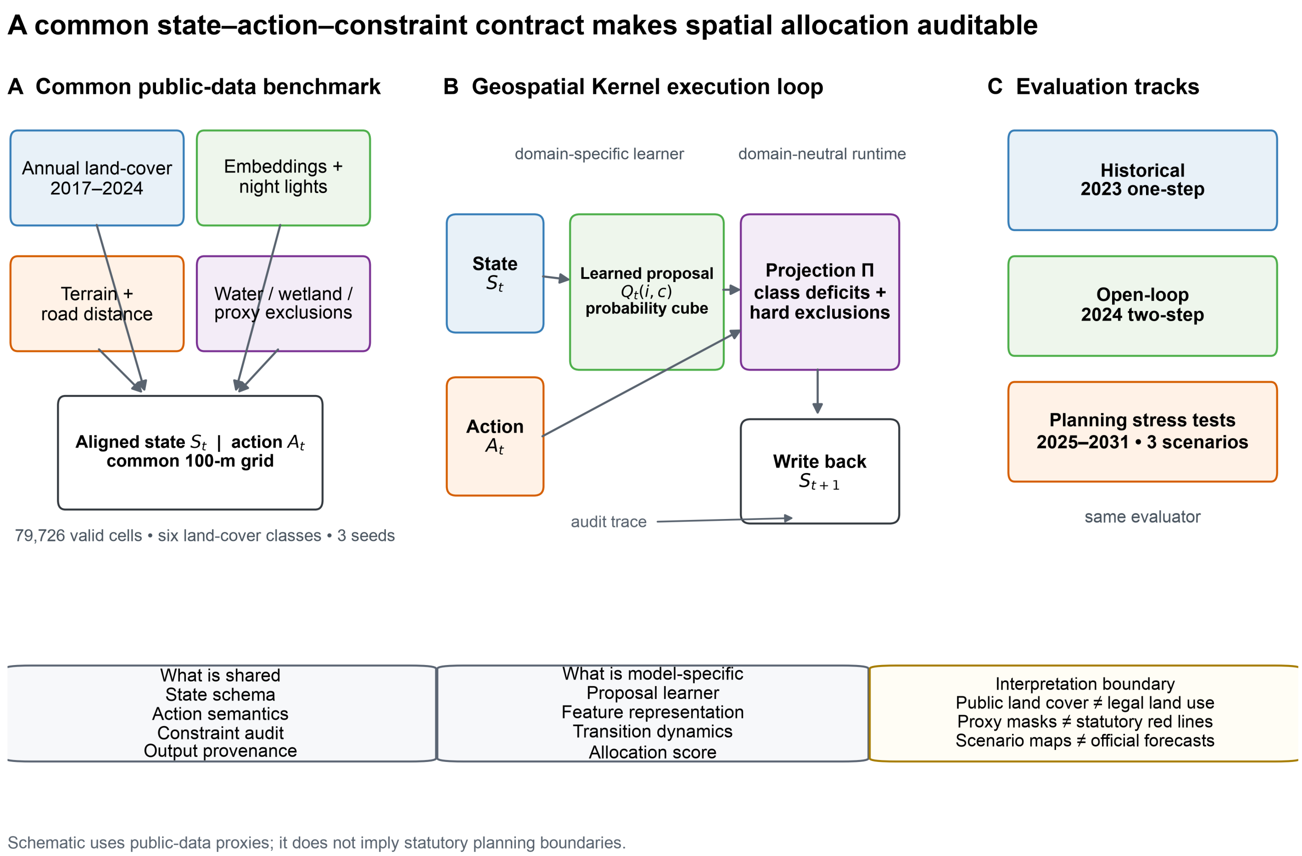

Figure 1 | Benchmark and Geospatial Kernel execution contract. Public annual land-cover, embedding, night-light, terrain and road layers are aligned to a common 100-m Abu Dhabi grid. Each model receives the same state, action and public proxy exclusions. The Geospatial Kernel exposes a learned proposal, constraint projection and state writeback. Historical, open-loop and planning tracks use one audit contract. The schematic does not imply statutory planning boundaries.

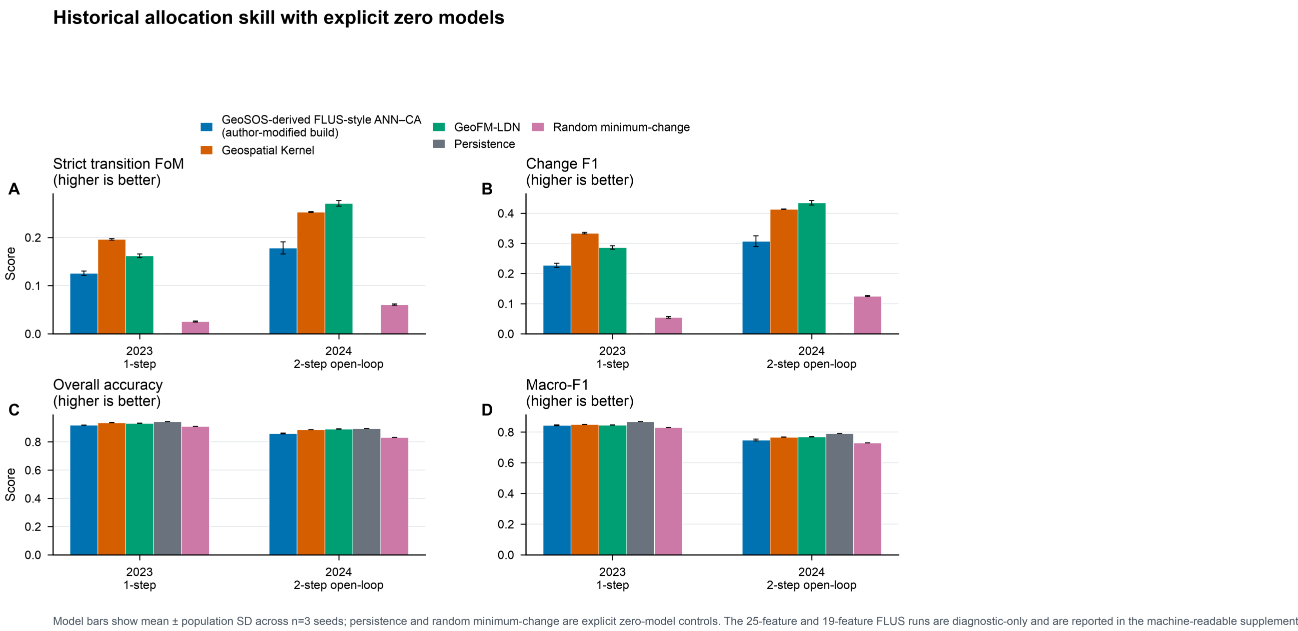

Figure 2 | Historical allocation skill. Bars show the revised strict multi-class FoM, change F1, overall accuracy and macro-F1 for the 2023 one-step and 2024 two-step open-loop tests, with persistence and random minimum-change controls. Error bars show population standard deviation across the three frozen seeds for the three main models. The 13-, 19- and 25-feature FLUS diagnostics are reported separately in Supplementary Table S3. Paired spatial-block intervals for the main models are reported in the machine-readable comparison report.

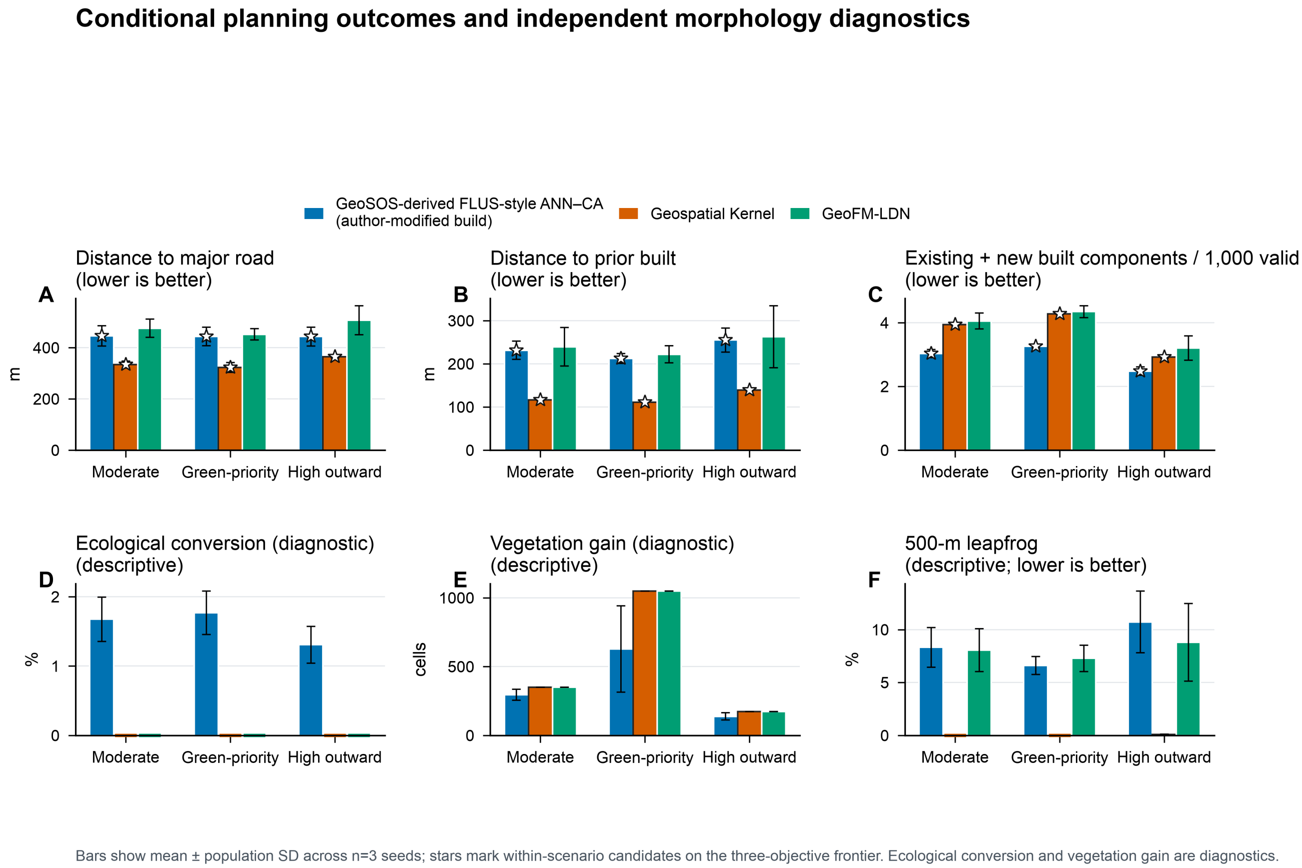

Figure 3 | Conditional planning objectives and morphology diagnostics. The 2031 outcomes compare the three models under moderate-growth, green-priority- growth and high-outward-growth actions. Panels A–C show the release objectives: distance to major roads, distance to prior built cells and components in the combined 2024-built and newly built footprint per 1,000 valid cells. Panel D shows ecological conversion as a descriptive diagnostic; panel E shows vegetation gain as a scenario-input diagnostic; panel F shows the 500-m leapfrog rate. Stars mark non-dominated candidates within each scenario; bars show means ± population SD across three seeds.

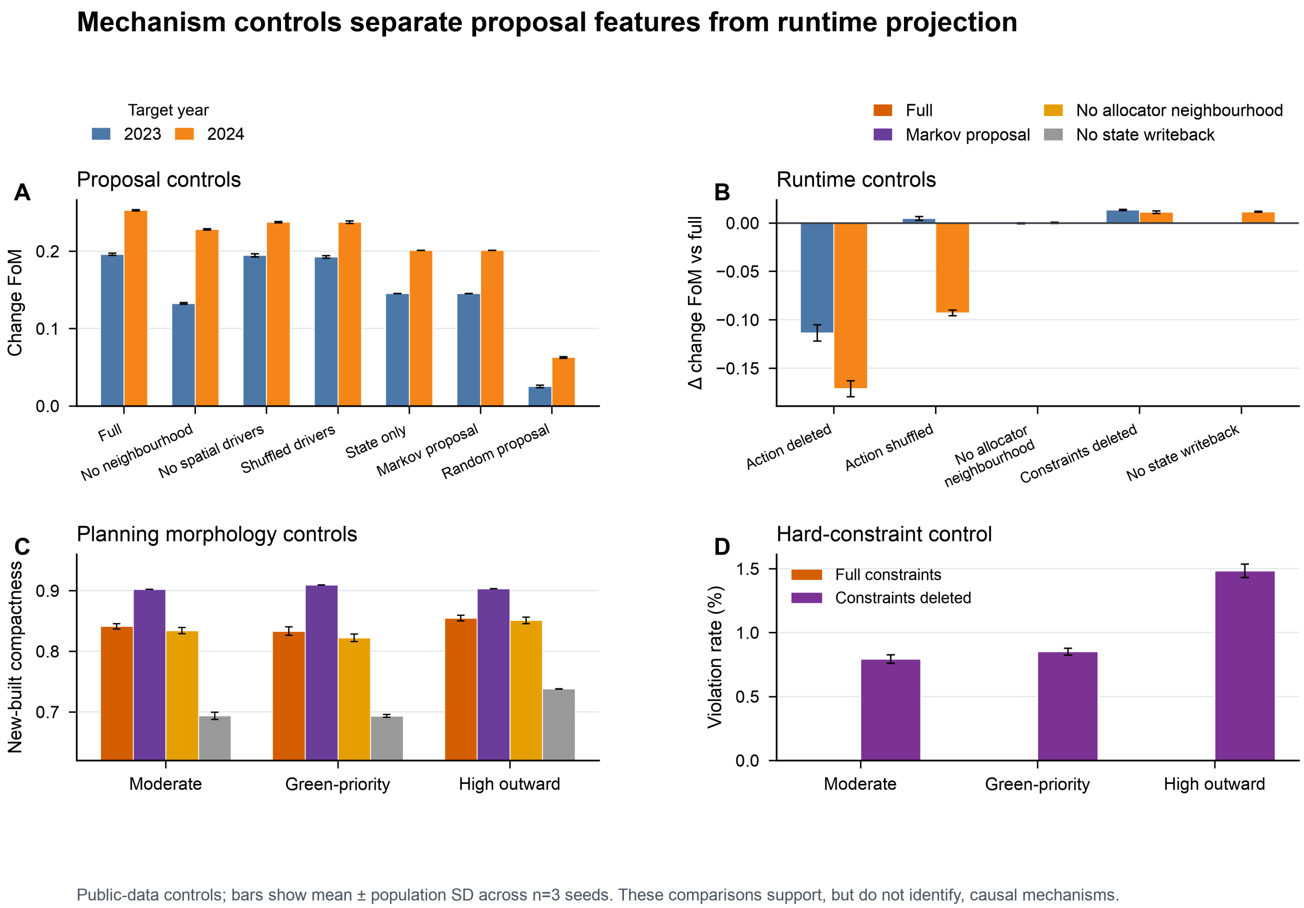

Figure 4 | Mechanism controls. Proposal, allocator, action, constraint and writeback controls are compared under the same public-data protocol. Deleting the action lowers historical strict FoM, whereas deleting the hard constraint raises agreement with noisy labels while violating the planning contract. These are diagnostic ablations, not causal policy experiments.

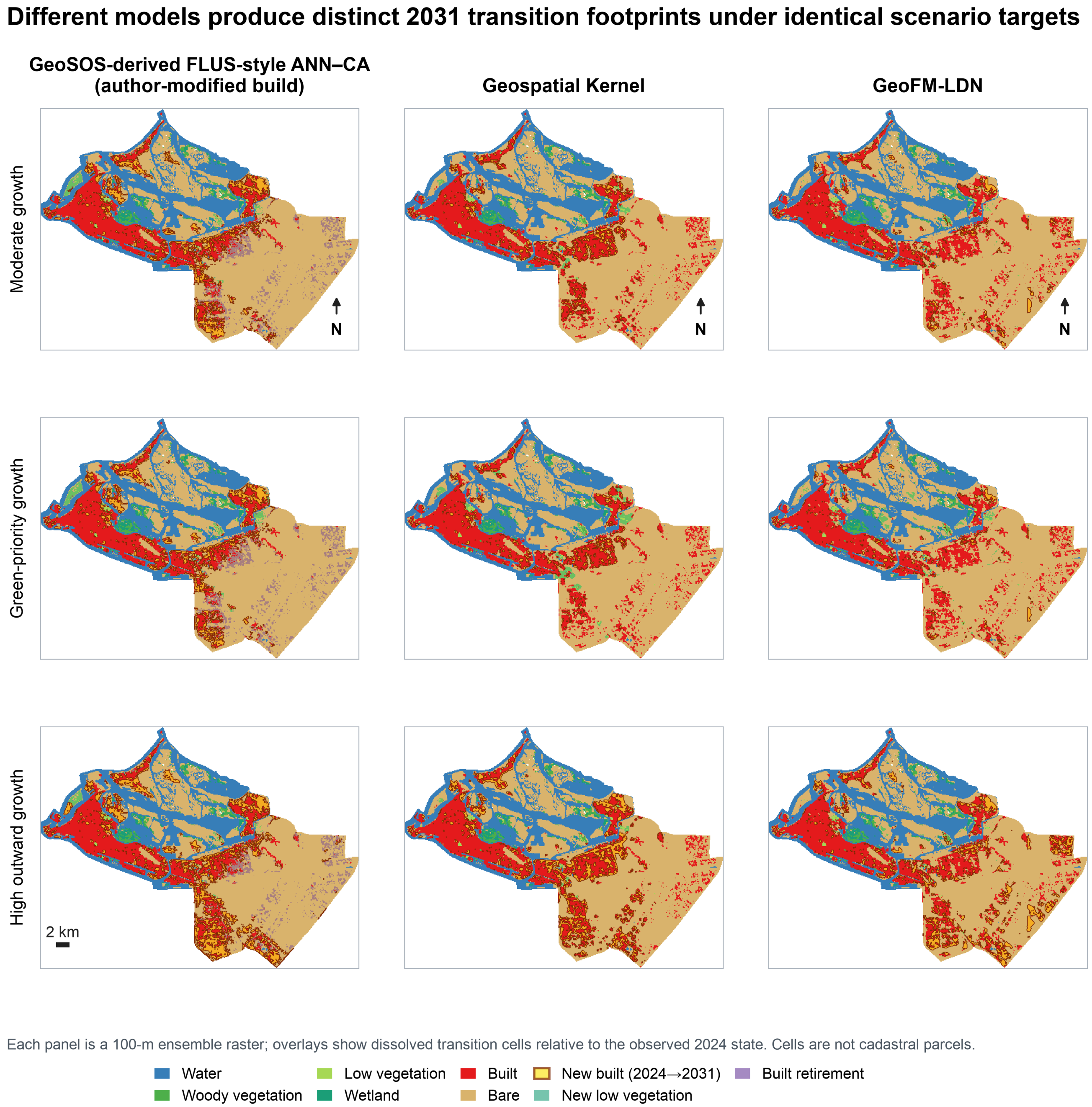

Figure 5 | 2031 spatial footprints. Each panel is a 100-m ensemble raster relative to the observed 2024 state. Overlays identify newly built, newly low- vegetated and retired-built cells; they are raster-cell change footprints, not cadastral parcels.

Supplementary information outline

- **Supplementary Note 1:** Complete data crosswalk, quality metrics and public/proxy status.
- **Supplementary Note 2:** Kernel runtime schemas, audit fields and adapter boundary.
- **Supplementary Table S1:** Full 2025–2031 model–scenario objective matrix with per-seed values.
- **Supplementary Table S2:** Neighbourhood-weight sensitivity at 0, 0.175, 0.35 and 0.7 (see `supplementary_table_S2_neighbourhood_weight_sensitivity.md`).

- **Supplementary Table S3:** FLUS 13-, 19- and 25-feature diagnostics, platform boundary and demand-underfill evidence (see `supplementary_table_S3_flus_feature_diagnostics.md`).
- **Supplementary Table S4:** Raster and vector delivery audit, hashes and layer counts.
- **Supplementary Figure S1:** Dynamic World confidence and high-confidence change sensitivity.
- **Supplementary Figure S2:** Historical maps and change-error maps for 2024.
- **Supplementary Figure S3:** Driver layers and experiment design in English.
